

QUIZ Sec F - 8.1 & 8.2

Q1. Define a relation S on Z as follows: For all integers a and b

$a S b \Leftrightarrow 3 \mid (a^3 - b^3)$ **(8 marks)**

a. Is $1 S 4$?

b. Is $2 S (-1)$?

c. Is $3 S 10$?

d. Is $(-5) S 5$?

Q2. Let $A = \{0, 1, 2, 3\}$. For each relation below:

1. Draw the directed graph. **(5 x 3 = 15)**

2. Determine if it is Reflexive, Symmetric, or Transitive. **(3 x 3 = 9)**

3. Provide counterexamples for any properties that fail. **(3)**

1. $R_1 = \{(0, 0), (1, 1), (2, 2), (3, 3), (0, 2), (2, 0)\}$

2. $R_2 = \{(0, 1), (1, 2), (0, 2)\}$

3. $R_3 = \{(0, 0), (0, 1), (1, 1), (2, 2)\}$